SED-DETR: A Scale-Enhanced Deformable Detection Transformer for Remote Sensing Images

Haitao Yin, *Member, IEEE*, Zhuyun Zhu, and He Wang

***Abstract*— Detection transformer (DETR) has emerged as a highly promising approach in object detection (OD) and has attracted significant interest. However, most DETR-like methods cannot simultaneously leverage the shape and scale priors for attention calculation, resulting in limited performance in detecting remote sensing objects with diverse shapes and scales. To address this issue, this article proposes a scale-enhanced deformable DETR (SED-DETR) for remote sensing OD (RSOD). The core component of SED-DETR is SED attention (SEDA), which is designed based on the principles of deformable shape and dynamic scale. Specifically, the SEDA module utilizes multiscale attention heads. First, conventional multiple attention heads are consolidated into several scale heads through an adaptive scale aggregation (Scale-Agg) approach, which dynamically adjusts the distributions of different scales to enhance the scale-aware modeling ability. For each scale head, dilated sampling is applied at a specific dilation rate to capture multiscale receptive fields. The sampled positions are further refined by learnable offsets predicted from query features, enabling a deformable dilated mechanism for fine-grained feature extraction of multiscale instances. Finally, we adopt the mixed query selection and the denoising training defined in DINO to implement SED-DETR. Experimental results on the xView, DIOR, NWPU VHR-10, and COCO datasets demonstrate that SED-DETR outperforms state-of-the-art (SOTA) DETR-like methods. Specifically, SED-DETR achieves 5.6%, 10.9%, and 8.6% mAP gains over the baseline deformable DETR on the xView, DIOR, and NWPU VHR-10 datasets, respectively. The source code is available at https://github.com/zzy599/SEDDETR**

***Index Terms*— Detection transformer (DETR), dilated sampling, position offsets, remote sensing object detection (RSOD), scale aggregation (Scale-Agg).**

I. Introduction

REMOTE sensing object detection (RSOD) aims to locate and categorize the objects of interest in remote sensing images, which is crucial to the intelligent interpretation of remote sensing in applications, such as environmental monitoring, mineral resource exploration, and urban planning [1], [2]. In contrast to natural images, remote sensing images are acquired from a bird's-eye view of the Earth's surface, covering wider regions and containing more objects. Moreover, remote sensing objects exhibit large variations in scale, shape, and orientation, posing unique challenges for object detection (OD). These complexities necessitate the RSOD approach with strong multiscale modeling and geometric adaptability.

In the deep learning era, the convolutional neural network (CNN) is a mainstream foundational model for image feature representation, which makes great progress in OD. As a distinctive subfield of OD, RSOD has also achieved remarkable results using CNN by taking into account task-specific characteristics. To deal with scale variation in RSOD, multiscale approaches have been extensively studied, such as the multiscale feature integration [3], [4] and the feature pyramid network [5], [6], [7], [8], [9]. Another extensively studied approach addresses the orientation challenge in RSOD, primarily through rotated object representation, such as adding extra parameters to encode the angle of the bounding box [10], [11], [12], [13] and applying angle classification to improve the accuracy of angle regression [14], [15], [16]. To reduce the limitations of anchor box definitions, keypoint-based anchor-free methods have also been explored in RSOD [17], [18], [19]. However, conventional CNN-based RSOD methods remain constrained by their reliance on handcrafted components (such as anchor boxes and keypoints) and the requirement of nonmaximum suppression (NMS). Moreover, the basic convolution operation cannot provide an optimal global understanding of wide-coverage remote sensing images.

Recently, detection transformer (DETR) [20] has achieved advanced performance, which is a fully end-to-end detection framework based on the transformer. Specifically, it utilizes a set of object queries to probe the objects of interest and their positions, without the requirements of anchor box design and NMS. However, compared with CNN-based methods, the original DETR suffers from low training convergence. As a consequence, many solutions have been explored to reduce redundancy and accelerate training. Token selection provides an intuitive approach by choosing a subset of tokens for attention calculation, thereby reducing computational overhead, such as deformable DETR [21], sparse DETR [22], focus DETR [23], salience DETR [24], and lite DETR [25]. Another approach is to design delicate object queries with more explicable information, such as conditional DETR [26], DAB-DETR [27], DAC-DETR [28], anchor DETR [29],

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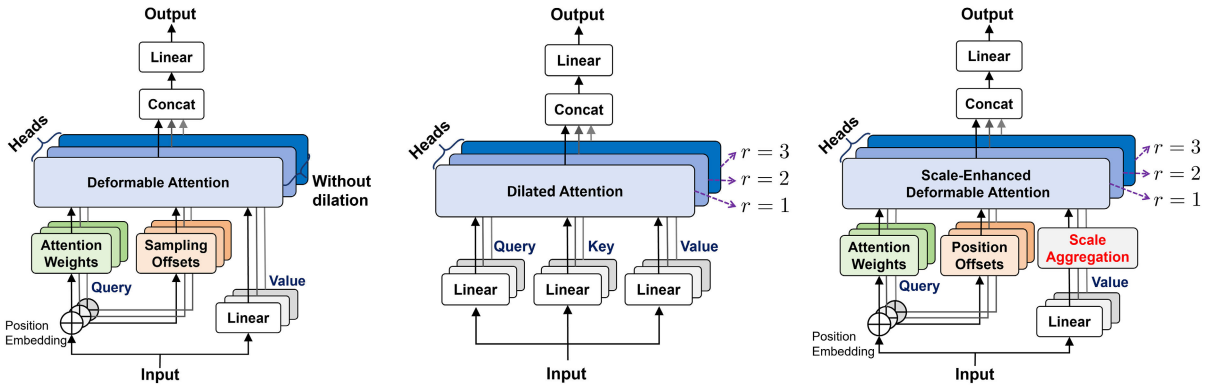


Fig. 1. Comparison between SEDA and related attention models. (a) DAT. (b) Dilated attention. (c) SEDA.

and enhancing DETR [30]. The third approach focuses on stabilizing the bipartite graph matching. For instance, DN-DETR [31] and DINO [32] incorporate a query denoising task during training. The classical one-to-one assignment is extended to one-to-many assignment [33], [34], [35], where each ground-truth object is assigned multiple predictions.

Inspired by the success of DETR, the remote sensing community has also applied DETR to RSOD and developed several variants. Most of these extensions aim to address the orientation issue, such as AO2-DETR [36], D^{2}Q-DETR [37], D-angle decoder [38], and ARS-DETR [39]. EMO2-DETR [40] is proposed to reduce the redundancy of queries in remote sensing images. To enhance interactions between objects, CI_DETR [41] replaces the feed-forward neural network (FFN) with capsule reasoning. To mitigate the domain gap, Han et al. [42] proposed a teacher–student DETR framework, incorporating a learnable frequency-enhanced feature alignment module. Despite these advances, the research on DETR-based RSOD remains scarce. Furthermore, the diverse scales and irregular shapes issues in RSOD have not been resolved effectively. Therefore, it is necessary and meaningful to improve the scale- and shape-aware abilities of DETR.

Based on the above considerations, this article proposes scale-enhanced deformable DETR (SED-DETR), a novel DETR framework for RSOD, which effectively overcomes the challenges of representing objects with diverse shapes and scales. Following the typical DETR framework, SED-DETR consists of a CNN backbone, a transformer encoder, and a transformer decoder. The fundamental difference between SED-DETR and the deformable attention (DAT)-based DETRs [21], [32] lies in that the SED attention (SEDA) is devised by exploiting the scale and shape priors for token selection. Specifically, SEDA adopts the deformable dilated sampling strategy to select points. First, we apply a scale aggregation (Scale-Agg) approach that consolidates conventional multiple attention heads into several scale heads. This dynamic scale approach can tackle the limited scalability issue of predefined scales. For each scale head, a fixed number of points centered at the position of each query is sampled with different dilation rates and then updated through a set of learnable offsets. Leveraging the refined dilated sampled points, the scale and shape priors are introduced into attention calculation, reducing redundancy, and accelerating training. The outputs of multiscale heads are integrated using concatenation and linear layers. Finally, we employ the mixed query selection and the contrastive denoising training in DINO [32] to initial queries and train SED-DETR, respectively. In addition, DINO is a DETR variant based on DAT. SED-DETR can also be regarded as an enhanced DINO, in which SEDA is utilized instead of DAT. A series of experiments on benchmark datasets demonstrate the superiority of SED-DETR. Fig. 1 shows the comparisons between SEDA and related attention models. SEDA has greater potential in understanding scale and shape than DAT [21] and dilated attention [43].

In summary, the contributions of this article are presented as follows.

1) We propose an SEDA module by jointly exploiting the interlevel/intralevel multiscale properties and shape adaptability, which provides a sound feature representation of diverse scales and irregular shapes for remote sensing objects.
2) Leveraging SEDA, we develop the SED-DETR for RSOD, which takes a full account of scale and shape priors to probe the objects of interest. It offers a feasible solution to low training convergence of DETR.
3) Extensive experimental results on the xVeiw, DIOR, NWPU VHR-10, and COCO datasets verify the superiority of SED-DETR compared with the state-of-the-art (SOTA) DETR-like methods.

The remainder of this article is organized as follows. In Section II, we briefly review the related works. Section III describes the proposed SED-DETR, including preliminaries, overall architecture, SEDA module, and some discussions. The experimental results and analyses are presented in Section IV. Finally, Section V concludes this article.

II. Related Work

A. Multiscale-Based RSOD Methods

The CNN-based RSOD methods originate from typical anchor-based detectors [44], [45], [46], [47], which are generally categorized as one-stage and two-stage. Due to the

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characteristics of remote sensing objects, numerous specific remote sensing detectors have been studied from the aspects of multiscale, rotation, weak small objects, and dense distribution. According to the goal of this article, we just briefly review the progress of multiscale methods.

The multiscale feature representation learning is crucial to the performance of multiscale methods. An intuitive approach is to integrate multiscale features of different levels of the CNN backbone and then perform a single prediction on the fused feature. Zhang et al. [3] proposed a hierarchical spatial semantic structure, which integrates hierarchical features, extracted from different layers of AlexNet. Zheng et al. [4] developed a fine-grained multiscale structure HyBlock through feature grouping, group connection, and aggregation. To reuse the semantic information of different layers, a single-shot multibox detector (SSD) [46] generates multiple predictions of pyramidal feature hierarchy, stacking multiple layers of CNN backbone. To improve the semantic information of low-level features in feature hierarchy, Lin et al. [48] proposed the feature pyramid network (FPN) with a top–down path. Since then, FPN has been widely used as the basic component of multiscale methods for RSOD, and its performance has also been improved gradually. For instance, Zhang et al. [5] proposed a Laplacian FPN, consisting of the bottom-up path, the Laplacian path, and the fusion path. Liu et al. [6] developed an adaptive FPN, which uses a selectively refined module to capture discriminative features. Liu et al. [17] designed a scene-enhanced FPN with a scene refine module to extract scene-level features. Gao et al. [7] developed a hidden recursive FPN, which encodes the relationship between before and after multiscale fusion. Zhang et al. [15] proposed a strengthened connection FPN, which extends the path aggregation network [49] through the lateral connection structure. To suppress background noise and highlight the information of objects, the attention module is incorporated into FPN, such as SFSANet [9], HCENet [50], and MAPFF [51]. In this article, we propose an alternative to the multiscale method through the DETR framework without handcrafted components.

B. Transformer-Based Detection Methods

As a counterpart to CNN, the transformer has also been applied to OD and has achieved dominant performance [52]. One reasonable transformer-based detection framework involves using the transformer to learn feature representation. Subsequently, similar to traditional CNN-based detectors, the detection heads are used to predict detection results. This transformer detection framework has also attained remarkable accomplishments in RSOD. For instance, CLT-Det [53] improves the performance for dense objects by learning the correlation between objects through the transformer. PSD-SQ [54] designs the transformer-based SQG module to express the semantic correlation among feature points and generate the corresponding query vectors. To improve the performance of local–global feature representation, Zhou et al. [55] developed a hybrid transformer and CNN framework.

DETR [20] is a new trend of transformer-based detection framework without anchor generation and NMS. Numerous follow-up works have been developed to speed up training and boost performance from the aspects of token selection, query design, and training schemes.

1) The first approach is to reduce redundancy by selecting informative tokens with specific criteria. Deformable DETR [21] uses DAT that selects a small set of points using the idea of deformable convolution. Sparse DETR [22] applies an encoder token sparsification algorithm to select and update a small part of tokens. Focus DETR [23] designs a scoring mechanism to determine the semantic level of tokens by considering both position and semantic information. Salience DETR [24] designs hierarchical query filtering refinement for selecting discriminative queries through salience-guided supervision. Lite DETR [25] applies an interleaved way to update high-level features and low-level features and then reduces the number of feature tokens.
2) The second approach aims to improve the informativeness of object queries. Conditional DETR [26] learns a conditional spatial query for cross-attention in the decoder, which is able to improve localization capability and accelerate training. DAB-DETR [27] uses the box coordinates as queries in the decoder, which are updated layer-by-layer. DAC-DETR [28] develops a "Divide-and-Conquer" decoder to avoid the competition between self-attention and cross-attention. Anchor DETR [29] designs object queries based on anchor points. Enhancing DETR [30] generates content queries for encoder embedding through self-attention pooling and uses a query aggregation strategy to improve the stability of one-to-one matching.
3) The third approach attempts to boost the stability of training through a denoising scheme and one-to-many assignment. DN-DETR [31] and DINO [32] introduce noises into ground-truth objects and subsequently perform denoising training. Group DETR [33] carries out the one-to-many assignment by conducting the one-to-one assignment for each group of object queries. MS-DETR [34] applies the mixed one-to-one assignment and one-to-many assignment for the primary decoder. Hybrid DETR [35] adds one additional decoder branch, in which the one-to-many assignment is executed.

The universal DETR framework has also been extended into RSOD. Considering the orientation information of remote sensing objects, various oriented DETR-like methods have been developed. AO2-DETR [36] uses oriented proposal generation, adaptive-oriented proposal refinement, and rotation-aware set matching loss to address the issues of misalignment, cluttered features, and limited matching. D^2Q-DETR [37] proposes to predict representative points for each rotated box instead of box regression. Zeng et al. [38] proposed a D-angle decoder by incorporating the position and angle information of objects into queries. ARS-DETR [39] uses a rotated DAT. To reduce the redundancy caused by a huge number of small remote sensing objects, Hu et al. [40] proposed the EMO2-DETR, consisting of reassigned bipartite

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graph matching, ignored sample prediction head, and reassigned Hungarian loss, which are used to extract and predict high-quality negative samples. Han et al. [41] designed the capsule reasoning module by using relative position information and category associations. Han et al. [42] introduced the DETR framework into unsupervised domain adaptive RSOD. In this article, we propose a novel DETR-like method for RSOD, which can alleviate the influences of diverse scales and irregular shapes.

III. Proposed Method

A. Preliminaries

DETR is a fully end-to-end anchor-free transformer-based OD method. The basic DETR framework consists of a CNN backbone, a transformer encoder, a transformer decoder, and multiple prediction heads. Given an input image, the CNN backbone first extracts the CNN features, which are then refined further through the transformer encoder. Subsequently, the encoded features and a set of learnable object queries are imported into the transformer decoder. The transformer decoder is composed of several stacked decoder blocks. Within these blocks, the object queries are updated gradually. The output of the last decoder block generates the final object queries, which are used to predict object class and box position within each prediction head through an FFN. Specifically, the bounding box is denoted as $\boldsymbol{b} = (b_x, b_y, b_w, b_h)$, where (b_x, b_y) are the center coordinates of the box, and b_w and b_h stand for the width and height of the box, respectively.

Self-attention is the core of the transformer. It formulates the interactions among all entities within the input sequence, thereby enabling the representation of global correlations between different positions. Let $X = (x_1, x_2, \ldots, x_N) \in \mathbb{R}^{N\times C}$ be an input sequence with N entities, where C is the dimension of each entity. The self-attention on X is defined as follows:

$$\mathrm{SA}(W_Q, W_K, W_V, X) = \sigma\left(\frac{XW_Q(XW_K)^\top}{\sqrt{d}}\right)XW_V \tag{1}$$

where σ denotes the softmax function, and $W_Q \in \mathbb{R}^{C\times d}$, $W_K \in \mathbb{R}^{C\times d}$, and $W_V \in \mathbb{R}^{C\times d}$ denote the linear projections for the query matrix, the key matrix, and the value matrix, respectively. Generally, the multihead self-attention (MHSA) is used to improve the performance of vanilla self-attention, that is, running M self-attention blocks in parallel and then concatenating with projection. Let $\{W_{Q_m}, W_{K_m}, W_{V_m}\}(m = 1, 2, \ldots, M)$ be the linear projections for the mth head. Then, the calculations of MHSA are given by

$$\begin{cases} \mathrm{Head}_m = \mathrm{SA}\left(W_{Q_m}, W_{K_m}, W_{V_m}, X\right) \\ Z = \mathrm{Concat}(\mathrm{Head}_1, \ldots, \mathrm{Head}_\mathrm{M})W_O \end{cases} \tag{2}$$

where d is set to $d = C/M$, $W_O \in \mathbb{R}^{C\times C}$ is a linear projection, Concat denotes the concatenation operation, Head_m is the output of mth head, and $Z = (z_1, z_2, \ldots, z_N) \in \mathbb{R}^{N\times C}$ is the output of MHSA.

In fact, each entity, obtained by (2), is the weighted sum of all entities in the sequence, where the weights are determined by the attention weights. Let A_m be the attention weights for the mth head. It can be written as follows:

$$A_m = \sigma\left(\frac{XW_{Q_m}(XW_{K_m})^\top}{\sqrt{d}}\right). \tag{3}$$

Moreover, we split W_O into M blocks, that is, $W_O = (W_{O_1}, W_{O_2}, \ldots, W_{O_\mathrm{M}})$, where $W_{O_m} \in \mathbb{R}^{C\times d}$ is the part related to the mth head. Then, the ith output entity z_i can be formulated as the following weighted summation form:

$$z_i = \sum_{m=1}^{M}\left(\sum_{j=1}^{N} A_{m,i,j}x_j W_{V_m}\right)W_{O_m}. \tag{4}$$

Inspired by deformable convolution [56], deformable DETR [21] and DAT transformer [57] use the DAT instead of traditional attention to save computational cost and reduce the influences of irrelevant regions. Our method follows the research line of deformable DETR. Hence, we briefly revisit the DAT used in deformable DETR. The core idea of DAT is that only a small number of feature points, sampled along the shape of the object, are assigned to each query. Let $p \in \mathbb{R}^{H\times W\times 2}$ be a reference point grid on the input feature $X \in \mathbb{R}^{H\times W\times C}$. DAT extends (4) as the following deformable form:

$$z_i = \sum_{m=1}^{M}\left(\sum_{j\in\Omega_i} A_{m,i,j}X\left(\phi\left(p_i + \triangle p_{m,i,j}\right)\right)W_{V_m}\right)W_{O_m} \tag{5}$$

where p_i is normalized as $p_i \in [0, 1]^2$, Ω_i denotes the set of samples for the ith query, ϕ denotes the bilinear interpolation, $\triangle p_{m,i,j} \in \mathbb{R}^2$ denotes the position offsets, and $A_{m,i,j}$ stands for the attention weight between the ith query q_i and the jth key with constraint $\sum_{j\in\Omega_i} A_{m,i,j} = 1$. In [21], $A_{m,i,j}$ and $\triangle p_{m,i,j}$ are predicted through linear projection from the query q_i. In practice, the number of elements in Ω_i can be fixed, namely, $|\Omega_i| = N$.

Moreover, the single-scale DAT (5) can be extended into its multiscale version, which uses multiscale features as input. Let $X_l \in \mathbb{R}^{H_l\times W_l\times C}(l = 1, 2, \ldots, L)$ be the input multiscale feature, extracted from different levels of CNN backbone. A subscript l is added into the variables in (5) to index the scale. Then, the multiscale DAT can be formulated as follows:

$$z_i = \sum_{m=1}^{M}\left(\sum_{l=1}^{L}\sum_{j\in\Omega_i} A_{m,l,i,j}X_l\left(\phi\left(\psi_l(p_i) + \triangle p_{m,l,i,j}\right)\right)W_{V_m}\right)W_{O_m} \tag{6}$$

where $\psi_l(p_i)$ rescales the coordinates of reference point for the lth scale. Empirically, multiscale DAT has a better performance than the single-scale model.

DAT gives a significant insight into shape prior to improving DETR. However, the current DAT has two limitations. First, the multiscale DAT (6) only considers the interlevel multiscale property, without exploiting the intralevel multiscale property within each feature map. Second, for the large object, the offsets $\{p_{m,l,i,j}\}_{m,l,i,j}$ should be generated using a long-range prediction with the constraint of a fixed number of samples, which could receive severe disturbances from the background

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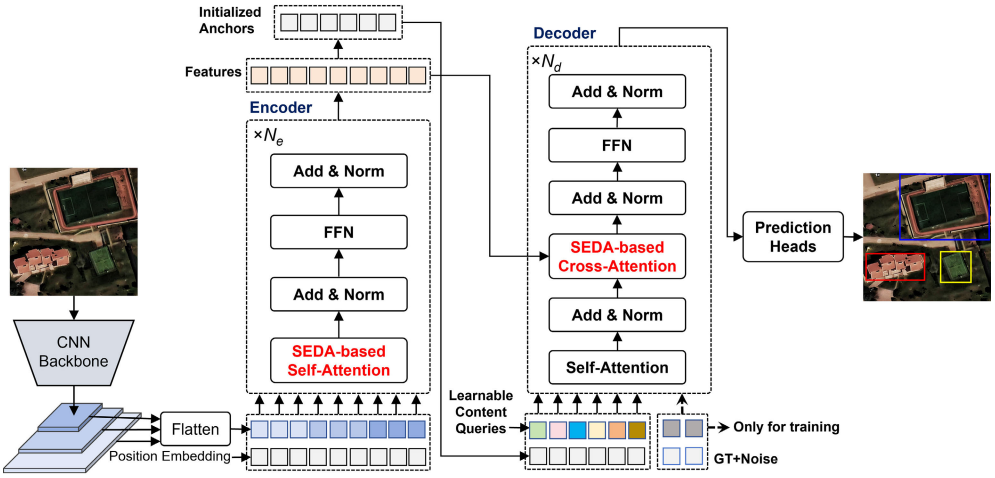


Fig. 2. Overall architecture of the proposed SED-DETR.

and other objects, resulting in inaccurate offsets generation. Motivated by these problems, our method jointly considers the interlevel and intralevel multiscale properties. In addition, a dynamic scale allocation strategy is used to enhance scale-aware adaptability.

B. Overall Architecture

Fig. 2 shows the overall architecture of SED-DETR. Following the architecture of typical DETR, we adopt the pretrained ResNet50 as the CNN backbone. Multiscale features extracted from different levels of the CNN backbone are flattened, and then, they are passed into the transformer encoder after adding the positional embeddings. Each encoder block adheres to the standard transformer block pipeline, but it replaces the conventional self-attention with the SEDA-based self-attention. In the decoder, we use the mixed query selection [32] to define the input queries. These queries include the initialized anchors (as positional queries) and the learnable content queries, which are then updated layer-by-layer. Each decoder block consists of typical self-attention and the SEDA-based cross-attention. The details of SEDA will be described in Section III-C. The updated object queries at the last decoder layer are used to predict the coordinates of the bounding box and class labels through multiple independent FFNs. Finally, SED-DETR is trained through the contrastive denoising training strategy [32].

C. SED Attention

To improve the awareness of instances with diverse scales and shapes, we propose the SEDA. Fig. 3 shows the diagram of SEDA, which is composed of Scale-Agg, dilated sampling, position update, attention calculation, and feature integration. The descriptions of these components are presented as follows.

1) Scale Aggregation: Let X be the input feature of SEDA. According to conventional MHSA, we generate the value $V = [V_1, V_2, \ldots, V_{\mathrm{M}}] \in \mathbb{R}^{M\times(C/M)\times H\times W}$ from X using linear projection, where $V_m \in \mathbb{R}^{(C/M)\times H\times W}$ is the mth ($m = 1, 2, \ldots, M$) head. In the conventional MHSA framework, all attention heads operate at the same scale. As a result, the intralevel multiscale property cannot be effectively exploited, which may lead to limited multiscale representation for each feature map. To alleviate this issue, we adopt the principle of dilated transformer [43], where each head samples feature points using a different dilation rate. However, the original dilated transformer fails to facilitate the interaction among different scales. Therefore, we develop a Scale-Agg approach, which consolidates M heads into $R (R < M)$ scale heads through dynamic selection weights $\{\pi_r\}_{r=1}^{R} \in \mathbb{R}^{M}$. Specifically, $\{\pi_r\}_{r=1}^{R}$ are predicted from the value through global average pooling (GAP), a linear layer, and softmax activation. The computations of $\{\pi_r\}_{r=1}^{R}$ are summarized as follows:

$$\{\pi_r\}_{r=1}^{R} = \sigma(\mathrm{Linear}(\mathrm{GAP}(\mathrm{Linear}(X)))). \tag{7}$$

Let $\widetilde{V}_r \in \mathbb{R}^{M\times(C/M)\times H\times W}$ be the value for the rth scale head. Then, $\widetilde{V}_r (r = 1, 2, \ldots, R)$ is produced by

$$\widetilde{V}^r = \pi_r \otimes V = \left[\pi_{r,1}V_1, \pi_{r,2}V_2, \ldots, \pi_{r,\mathrm{M}}V_{\mathrm{M}}\right] \tag{8}$$

where $\pi_{r,m}$ is the mth element of π_r. Subsequently, the SEDA is applied to these R scale heads $\{\widetilde{V}^r\}_{r=1}^{R}$.

2) Dilated Sampling: For different scale heads, we employ different dilation rates to sample the entries of the value for the target query point. Without loss of generality, the dilation rate r is assigned to the rth head ($r = 1, 2, \ldots, R$). Given the query position $p_i \in \mathbb{R}^2$ of the ith query, the value sampling strategies for different scale heads are given as follows.

a) For $r = 1$, the value points are sampled within a standard $k \times k$ window centered at p_i.
b) For $r \geq 2$, the sampling positions are determined by applying dilation rate r to the sampling grid.

In short, the dilated sampling set $\Gamma_{r,i}$ for query point $q_{r,i}$ at the rth scale head is given by

$$\Gamma_{r,i} = \left\{ p_{r,j} \mid p_{r,j} = p_i + r \cdot w, w \in \left[-\frac{k}{2}, \frac{k}{2}\right]^2 \right\}. \tag{9}$$

In addition, at the boundary of the feature map, the zero padding is used to assist the dilated sampling in (9), as shown in the second head in Fig. 3.

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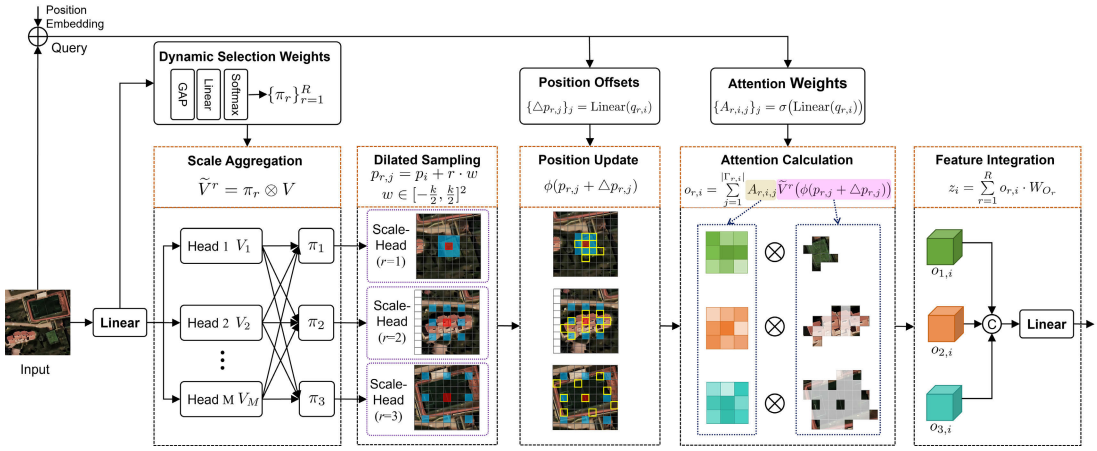


Fig. 3. Illustration of SEDA.

3) Position Update: Indeed, the strategy in (9) indicates that feature points are sampled within a regular $k_r \times k_r$ window for the rth head, where $k_r = k + (k-1)(r-1)$. Although basic dilated sampling enlarges receptive fields, it lacks the flexibility to accurately model instance shapes. To remedy this limitation, we adopt a set of learnable position offsets to adaptively update sampled positions. Thus, the sampled positions are updated as follows:

$$\widetilde{\Gamma}_{r,i} = \left\{\phi\left(p_{r,j} + \triangle p_{r,j}\right),\ \ p_{r,j} \in \Gamma_{r,i}\right\} \tag{10}$$

where $\triangle p_{r,j} \in \mathbb{R}^2$ denotes the offset for the position $p_{r,j}$ and ϕ denotes the bilinear interpolation. Following DAT, we also normalize the positions $p_{r,j}$ into $[0, 1]$. The offset $\triangle p_{r,j} \in [0, 1]$ is predicted from the ith query through a linear layer, that is, $\{\triangle p_{r,j}\}_j = \text{Linear}(q_{r,i})$. Hence, for the ith query at the position $p_{r,i}$, the value points are selected from the set $\widetilde{\Gamma}_{r,i}$ to calculate attention at the rth scale head.

4) Attention Calculation: Based on (8)–(10), the jth value point for the ith query at the rth scale head is $\widetilde{V}^r\left(\phi(p_{r,j} + \triangle p_{r,j})\right)$, where $p_{r,j} \in \Gamma_{r,i}$. Instead of dot-product similarity in typical self-attention, we propose to predict the attention weights from the query through a linear layer and softmax function. Let $A_{r,i,j}$ be the attention weight between the ith query and the jth key at the rth scale head. $\{A_{r,i,j}\}_{j=1}^{|\Gamma_{r,i}|}$ is obtained by

$$\{A_{r,i,j}\}_{j=1}^{|\Gamma_{r,i}|} = \sigma\left(\text{Linear}(q_{r,i})\right) \tag{11}$$

where $|\Gamma_{r,i}|$ computes the power of set $\Gamma_{r,i}$. Then, the attention of the rth scale head is calculated as follows:

$$o_{r,i} = \sum_{j=1}^{|\Gamma_{r,i}|} A_{r,i,j}\, \widetilde{V}^r\left(\phi(p_{r,j} + \triangle p_{r,j})\right) \tag{12}$$

where $o_{r,i}$ denotes the ith element in the output of the rth scale head. As seen in (12), every scale head in SEDA operates on the complete input feature X. This is different from typical MHSA, in which each head operates on a partition of input features. Therefore, the dilation rate defined in (9) can be set to any value without the dimension mismatch issue that may occur in the typical MHSA.

5) Feature Integration: The final output of SEDA is produced through feature integration of all scale heads. Specifically, it is operated through a concatenation operation followed by a linear layer W_O, which is formulated as follows:

$$z_i = \text{Concat}\left(o_{1,i}, o_{2,i}, \ldots, o_{R,i}\right) \cdot W_O = \sum_{r=1}^{R} o_{r,i} \cdot W_{O_r} \tag{13}$$

where $W_O = [W_{O_1}, W_{O_2}, \ldots, W_{O_R}]$ is the split of W_O according to the dimensions of $\{o_{r,i}\}_{r=1}^{R}$.

6) Multiscale Implementation and Whole Formulation: The proposed SEDA can be implemented using the multiscale features X_l $(l = 1, 2, \ldots, L)$ as input. We introduce the subscript l to index the lth scale, and the aforementioned considerations in the single-scale version are merged into one framework. Finally, the whole formulation of multiscale SEDA is defined as follows:

$$z_i = \sum_{r=1}^{R}\left(\sum_{l=1}^{L}\sum_{j=1}^{|\Gamma_{r,i}|} A_{r,l,i,j}\, \widetilde{V}^r\left(\phi(\psi_l(p_{r,j}) + \triangle p_{r,l,j})\right)\right) \cdot W_{O_r} \tag{14}$$

where ψ_l has the same definition in (6), and the attention weights $A_{s,l,i,j}$ satisfy $\sum_{l=1}^{L}\sum_{j=1}^{|\Gamma_{r,i}|} A_{r,l,i,j} = 1$.

Recalling the overall architecture in Section III-B, the SEDA is equipped with the self-attention of the encoder and the cross-attention of the decoder. The difference between these two arrangements of SEDA is that the queries are different. Specifically, the queries in the self-attention of the encoder are from image features, while the queries in the cross-attention of the decoder derive from object queries.

D. Discussion

Table I summarizes the differences between MSDA [43], DAT [21], and SEDA from different perspectives. It can be observed that SEDA provides a more comprehensive understanding of shape and scale properties than MSDA and DAT, which facilitates the detection of instances with various scales. Furthermore, the combination of dilated sampling and position update can be regarded as deformable dilated sampling. In contrast to the deformable sampling in DAT, the deformable

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TABLE I
PROPERTIES OF DIFFERENT ATTENTION MODELS

Types	Inter-Level Multi-Scale	Intra-Level Multi-Scale	Dynamical Scale	Deformable Shape
MSDA [43]	×	✓	×	×
DAT [21]	✓	×	×	✓
SEDA (Ours)	✓	✓	✓	✓

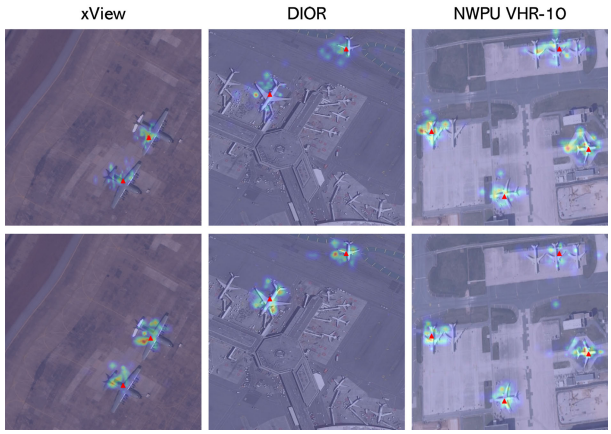


Fig. 4. Visualizations of the attention maps generated by (top row) DAT and (bottom row) SEDA. The red triangles and the highlighted regions indicate the query of interest and the attention regions, respectively.

dilated sampling involves a shorter range prediction of shape, which enables such sampling to be more accurate and robust.

Fig. 4 presents the visualization of the attention maps of the last encoder block, which are derived from DAT (in the DINO) and SEDA. In particular, the attention maps not only display the sampling points but also reveal the underlying dependencies. Compared with the attention maps generated by DAT, the attention maps generated by SEDA are more congruent with the shapes of aircraft (xView dataset), more accurate for small aircraft (DIOR dataset), and more discriminative for densely distributed aircraft (NWPU VHR-10 dataset).

IV. EXPERIMENTS

A. Experimental Setting

1) Datasets: We conduct the experiments on three remote sensing image datasets (xView [63], DIOR [64], and NWPU VHR-10 [65]) and one natural image dataset (COCO 2017 [66]). The brief descriptions of these datasets are presented as follows.

a) The xView dataset derives from the DIUx xView 2018 detection challenge, which is one of the largest publicly available remote sensing datasets. The images are acquired by the WorldView3 satellite with a ground sampling range resolution of 0.3 m, spanning a total area of over 1400 km^2, and containing more than one million object instances with 60 object categories. We follow the processing strategy in [67] to process the dataset. Since the original validation set has no labels, 846 training images are randomly split into 700 images for training and the remaining 146 images for validation.

b) The DIOR dataset was constructed by Northwestern Polytechnical University in 2020. It contains 23 463 images and more than 190 000 instances with resolution ranging from 0.5 to 30 m, which contains 20 object categories, including airplane (AL), airport (AO), baseball field (BF), basketball court (BC), bridge (BD), chimney (CH), expressway service area (ES), expressway toll station (ET), dam (DM), golf field (GL), ground track field (GT), harbor (HB), overpass (OP), ship (SH), stadium (SD), storage tank (ST), tennis court (TC), train station (TS), vehicle (VH), and windmill (WM). Each image is with the size of 800 × 800. There are 5862 images in the training set, 5862 images in the validation set, and 11 738 images in the test set. We use the training set and the validation set for the training model and performance evaluation, respectively.

c) The NWPU VHR-10 dataset is a high-resolution remote sensing dataset, which contains 800 images with a size of 512 × 512, a total of 3651 objects, and ten object categories, including AL, ship (SH), ST, baseball diamond (BD), TC, BC, GT, harbor (HB), bridge (BR), and vehicle (VH). There are 650 images containing one or more objects as positive images and 150 images without objects as negative images. Following dataset processing in [61], we randomly divide the 650 positive images into 520 images as the training set and the remaining 130 images as the validation set.

d) The COCO 2017 dataset is a benchmark natural image dataset, containing 80 object categories in total. The training and evaluation are conducted on the `train2017` split (118 287 images) and the `val2017` split (5000 images), respectively.

2) Implementation Details: For a fair comparison, the training settings of SED-DETR follow the settings in DINO [32]. The pretrained ResNet50 on the ImageNet-1K dataset is used as the CNN backbone. The layers of encoder and decoder are all set to 6, namely, $N_e = 6$ and $N_d = 6$ in Fig. 2. The dimension of hidden features is chosen as 256. We use 900 object queries. The window size k and the head number D are both set to 3. The AdamW optimizer is used to train model parameters. We employ a two-stage training strategy to train our model. First, we train the model on the COCO dataset for 12 epochs and save it as the pretrained model. Subsequently, the pretrained model is fine-tuned on the remote sensing dataset for another 12 epochs. For each training stage, the initial learning rates for the CNN backbone and other modules are set to 10^{-5} and 10^{-4}, respectively. Subsequently, both learning rates are reduced by a factor of 0.5 at the 11th epoch. The batch size in the two training stages is set to 8. The other DETR-like methods are trained in the

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TABLE II

COMPARISON RESULTS [AP(%)] WITH VARIOUS DETECTION METHODS ON THE XVIEW DATASET. THE AP RESULTS ARE CALCULATED FOLLOWING THE DEFINITIONS IN THE PASCAL VOC DATASET [58]. THE RESULTS OF 20 SEPARATE SMALL OBJECT CATEGORIES ARE SELECTED AND PRESENTED

Categories	YOLOv8 [59]	GLIP [60]	SuperYOLO [61]	YOLOv10 [62]	Deformable DETR [21]	DN-DETR [31]	DINO (4scale) [32]	DINO (5scale) [32]	SED-DETR (4scale)
Small Car	78.7	75.7	72.5	75.5	70.0	60.1	76.7	77.8	70.5
Bus	35.0	32.8	37.4	40.8	33.4	27.4	41.4	39.4	45.8
Truck	19.4	18.6	19.6	18.7	16.4	18.1	21.1	24.8	22.7
Cargo Truck	11.7	10.3	12.5	15.6	12.1	9.4	17.4	14.7	16.9
Trailer	18.5	19.2	15.3	18.2	16.8	11.5	19.4	21.4	25.4
Cargo Car	6.9	7.3	9.9	14.3	11.0	9.5	7.1	8.1	15.8
Maritime Vessel	23.1	20.7	19.7	21.7	20.0	23.8	27.7	23.6	29.5
Motorboat	45.3	50.1	49.6	51.6	55.1	37.4	53.3	57.9	59.9
Sailboat	8.7	7.7	14.8	17.7	24.2	12.5	6.4	22.8	23.0
Tugboat	21.2	19.6	20.0	27.6	16.0	30.2	29.9	26.8	27.6
Barge	16.1	10.5	7.9	10.1	10.5	9.0	18.4	16.0	25.3
Fishing Vessel	16.3	11.4	16.5	19.4	21.5	13.2	17.6	22.2	37.0
Ferry	20.7	16.7	3.8	6.8	7.0	7.2	26.6	16.1	20.7
Yacht	29.6	28.4	23.3	36.3	31.1	45.7	36.8	39.7	54.3
Container Ship	44.4	37.8	38.5	39.8	48.5	48.8	53.3	43.1	57.3
Straddle Carrier	38.7	36.9	29.1	36.8	23.2	40.0	65.4	29.1	65.7
Mobile Crane	9.5	9.4	7.5	11.4	7.4	13.3	13.3	17.7	14.0
Front loader/Bulldozer	30.6	30.5	30.5	33.5	30.6	34.7	46.2	40.9	37.4
Excavator	51.2	48.3	55.3	58.3	60.5	58.3	65.7	60.2	61.6
Cement Mixer	7.6	5.6	6.6	9.6	20.9	10.5	13.1	20.9	50.6
mAP	22.9	20.7	20.1	23.3	22.0	21.2	27.1	26.0	**27.6**

same way. The experiments are performed on the NVIDIA L20 GPU.

3) Loss Function: Following DINO [32], we use the ℓ_1 loss and GIOU loss for bounding box regression, and the focus loss with $\alpha = 0.25$ and $\gamma = 2$ for object classification. In addition, the coefficients for ℓ_1 loss, GIOU loss, and focus loss are set to 5, 2, and 1, respectively.

4) Metrics: To quantify the detection performance, we adopt the average precision (AP) with different IoU thresholds. AP_{50} and AP_{75} denote the AP values with IoU thresholds greater than 0.5 and 0.75, respectively. AP is calculated with the IoU thresholds from 0.5 to 0.95 at intervals of 0.05. mAP is the mean AP of all categories. Apart from AP, AP_{50}, and AP_{75}, we also employ three variants of AP based on the size of object, that is, AP_S, AP_M, and AP_L are used to measure the performance of small, medium, and large objects, respectively. Specifically, the small, medium, and large objects mean that the sizes of objects are less than 32×32, between 32×32 and 96×96, and greater than 96×96, respectively. The AP metrics are calculated following the definitions in the COCO dataset and the Pascal VOC dataset [58] to meet different requirements.

B. Comparison on xView Dataset

Table II presents the quantitative comparison on the xView dataset between SED-DETR and other detectors, including the mAP results of all categories and the AP results of 20 separate small object categories (small car, bus, truck, cargo truck, trailer, cargo car, maritime vessel, motorboat, sailboat, tugboat, barge, fishing vessel, ferry, yacht, container ship, straddle carrier, mobile crane, front loader/bulldozer, excavator, and cement mixer). It can be seen that SED-DETR obtains the best score of mAP. Compared with the DAT-based deformable DETR and DINO (4scale), SED-DETR achieves mAP gains of 5.6% and 0.5% mAP, respectively. Fig. 5 visualizes the detection results of DINO and SED-DETR on the xView dataset. For clear visual comparison, one region of each result is selected and enlarged. As shown in the close-ups of the first image, DINO misclassifies the "small aircraft" instance as the "cargo plane" category, while SED-DETR provides the correct detection result. It can verify that SED-DETR can overcome the interclass similarity issue more effectively than DINO. For the other three images, DINO misses some instances, such as the container ship and small cars in the second image, the building in the third image, and the boat in the fourth image. In contrast, SED-DETR obtains lower false negatives.

C. Comparison on DIOR Dataset

Table III exhibits the qualitative comparison of SED-DETR with other methods on the DIOR dataset. It can be seen that SED-DETR achieves up to 80.3% mAP, which is 10.9% higher than deformable DETR and 0.1% higher than DINO (4scale). Fig. 6 shows the visual comparison between DINO and SED-DETR. In the first image, DINO treats the shadow of storage as the "storage," resulting in a false positive. By contrast, SED-DETR can reduce the influence of shadow (background) more effectively. Besides, the comparisons in the second, third, and fourth images can verify that SED-DETR generates better detection performance on small objects since DINO misses some small objects, such as the ships in the second image.

D. Comparison on NWPU VHR-10 Dataset

In this section, we give the performance evaluation of the NWPU VHR-10 dataset, which faces challenges in detecting dense objects. Table IV shows the mAP values of compared methods. SED-DETR achieves the highest detection accuracy, 8.6% higher than deformable DETR. Moreover, under the same settings, it offers 2.0% mAP gains over DINO (4scale). Fig. 7 displays the visualizations of four pairs of detection

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Fig. 5. Visualization of detection results on the xView dataset. (Top and bottom rows) Results of DINO (4scale) and SED-DETR, respectively.

TABLE III

COMPARISON RESULTS [AP(%)] WITH VARIOUS DETECTION METHODS ON DIOR DATASET. THE AP RESULTS ARE CALCULATED FOLLOWING THE DEFINITION IN THE PASCAL VOC DATASET [58]

Methods	AL	AO	BF	BC	BD	CH	ES	ET	DM	GL	GT	HB	OP	SH	SD	ST	TC	TS	VH	WM	mAP
YOLOv8 [59]	92.2	82.3	93.7	81.8	48.5	89.6	65.8	73.7	67.0	75.7	80.1	69.5	66.4	93.3	92.1	81.4	92.3	50.4	73.2	81.6	77.5
GLIP [60]	69.7	78.9	71.6	87.2	61.8	68.5	79.4	72.3	71.7	77.6	76.9	57.8	61.3	74.6	73.9	63.4	83.3	62.8	50.3	75.4	71.4
SuperYOLO [61]	81.2	57.3	83.6	89.5	41.6	77.0	71.9	76.2	40.1	50.0	69.0	48.1	44.8	89.5	76.0	76.0	91.3	33.7	50.9	89.2	66.9
YOLOv10 [62]	81.1	91.7	84.9	90.0	51.9	79.6	91.8	77.5	78.7	84.2	84.1	67.4	65.8	92.1	79.6	78.1	92.2	70.2	58.0	90.6	79.5
Deformable DETR [21]	68.9	81.5	74.6	84.3	46.0	78.9	79.4	73.2	64.5	71.4	79.6	43.7	59.2	75.7	66.2	63.4	87.4	58.2	50.3	80.8	69.4
DN-DETR [31]	74.0	85.3	77.7	87.4	51.7	80.1	88.2	77.2	72.5	82.9	82.5	45.3	62.8	69.4	73.7	62.5	87.8	68.4	51.3	89.9	73.5
DINO (4scale) [32]	87.7	90.8	85.6	91.1	57.6	84.8	93.8	81.1	79.3	86.2	86.2	64.2	69.5	78.9	80.0	73.7	91.8	70.1	61.0	91.5	80.2
DINO (5scale) [32]	81.4	89.3	85.1	89.9	56.0	83.8	91.3	84.0	80.8	85.8	86.6	61.1	67.5	78.1	76.6	74.6	92.2	72.3	61.0	91.0	79.4
SED-DETR (4scale)	85.0	92.5	86.3	91.0	55.5	84.6	92.8	84.0	77.2	86.5	86.0	67.1	67.8	79.8	80.5	74.5	92.9	71.7	59.0	90.4	**80.3**



Fig. 6. Visualization of detection results on the DIOR dataset. (Top and bottom rows) Results of DINO (4scale) and SED-DETR, respectively.

results. Through the comparisons in the first, third, and fourth pairs, we can obtain that SED-DETR produces more complete and more accurate detection of dense small objects than DNIO. Furthermore, the comparisons in the second pair show that the bounding boxes of SED-DETR are fitter to the shapes of objects. Overall, SED-DETR can effectively address the shape and scale issues of remote sensing objects.

E. Comparison on COCO Dateset

Finally, the performance comparison is conducted on the COCO dataset. Table V presents the quantitative comparison on COCO `val2017`. It can be observed that SED-DETR obtains the best results in terms of AP, AP_{50}, AP_{75}, AP_M, and AP_L. SED-DETR offers 0.4% AP gains over DINO (4scale). In addition, compared with deformable DETR, the AP gains increase to 3.2%. In comparison to other DETR variants, the gains of SED-DETR are also remarkable. Specifically, SED-DETR provides 0.2% AP improvements over the latest Salience DETR.

F. Ablation Studies

In this section, we present the ablation studies on major components of SED-DETR, including the number of scale

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TABLE IV

COMPARISON RESULTS [AP(%)] WITH VARIOUS DETECTION METHODS ON THE NWPU VHR-10 DATASET. THE AP RESULTS ARE CALCULATED FOLLOWING THE DEFINITION IN THE PASCAL VOC DATASET [58]

Methods	AL	SH	ST	BD	TC	BC	GT	HB	BR	VH	mAP
YOLOv8 [59]	99.1	88.2	96.0	98.8	89.4	70.1	96.6	84.1	83.2	80.0	88.6
GLIP [60]	90.9	72.7	89.3	89.9	90.0	88.3	98.8	81.2	97.0	82.7	88.1
SuperYOLO [61]	99.3	86.0	99.0	97.5	97.5	77.5	99.3	93.6	86.2	91.3	92.7
YOLOv10 [62]	98.2	91.3	98.1	97.9	95.6	73.5	96.3	79.1	66.5	87.7	88.4
Deformable DETR [21]	94.3	84.8	81.5	94.7	78.9	87.6	94.9	86.5	73.2	84.3	86.1
DN-DETR [31]	90.2	85.6	89.6	92.4	84.3	67.8	94.7	77.5	73.2	81.3	83.7
DINO (4scale) [32]	98.9	93.4	96.1	97.9	94.1	95.4	95.8	91.6	71.2	92.5	92.7
DINO (5scale) [32]	96.9	89.6	93.9	96.4	90.6	89.2	94.6	91.3	86.9	89.8	91.9
SED-DETR (4scale)	99.9	92.8	97.9	95.9	94.9	93.0	94.5	94.6	94.3	89.5	**94.7**

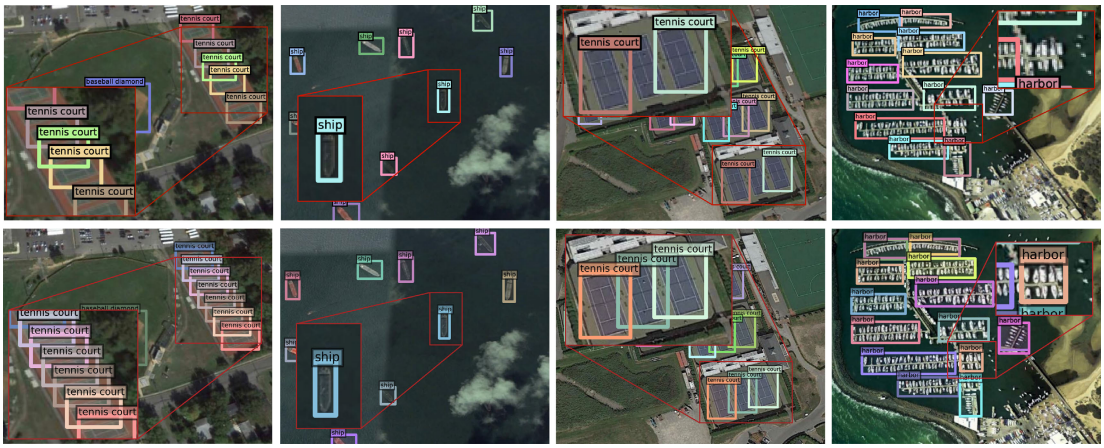


Fig. 7. Visualization of detection results on the NWPU VHR-10 dataset. (Top and bottom rows) Results of DINO (4scale) and SED-DETR, respectively.

TABLE V

COMPARISON RESULTS WITH VARIOUS DETECTION METHODS ON THE COCO DATASET. FOR THE MODELS WITHOUT MULTISCALE FEATURES, WE TEST THEIR PERFORMANCE WITH RESNET-50-DC5, DENOTED AS "−DC5." THE AP METRICS ARE CALCULATED FOLLOWING THE DEFINITIONS IN THE COCO DATASET [66]

Methods	Epochs	Backbone	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	#Params
Faster-RCNN [44]	108	ResNet50	42.0	62.4	44.2	20.5	45.8	61.6	40M
DETR-DC5 [20]	500	ResNet50	43.3	63.1	45.9	22.5	47.3	61.6	41M
Conditional DETR-DC5 [26]	108	ResNet50	45.1	65.4	48.5	25.3	49.0	62.2	44M
Deformable DETR [21]	50	ResNet50	46.2	65.2	50.0	28.8	49.2	61.7	40M
DAB-Deformable-DETR [27]	50	ResNet50	46.9	66.0	50.8	30.1	50.4	62.5	47M
DN-Deformable-DETR [31]	50	ResNet50	48.6	67.4	52.7	31.0	52.0	63.7	48M
Salience DETR [24]	12	ResNet50	49.2	67.1	53.8	**32.7**	53.0	63.1	56M
DINO (4scale) [32]	12	ResNet50	49.0	66.6	53.5	32.0	52.3	63.0	47M
DINO (5scale) [32]	12	ResNet50	**49.4**	67.0	53.8	32.3	52.5	63.9	47M
SED-DETR (4scale)	12	ResNet50	**49.4**	**67.6**	**54.1**	31.6	**53.2**	**65.5**	50M

heads, Scale-Agg, and SEDA. To conduct the ablation experiments, we built several variants of SED-DETR. It should be noted that all variants only modify the configurations of related components, while the other implementations and training strategies are kept fixed. The NWPU VHR-10 dataset is used as an experimental dataset. The AP metrics defined in the COCO dataset [66] are employed for performance evaluation.

1) Effect of Number of Scale Heads: In SED-DETR, the number of scale heads is denoted as R, which not only controls the number of heads but also determines the deformable dilated sampling. To study the effect of R, we carry out SED-DETR with different settings of R. We adopt the same rule to set the dilation rates, that is, the rth scale head uses the dilation rate r. Table VI shows that the "$R = 3$ with dilation rates $[1, 2, 3]$" version generates the best results in all AP metrics. Hence, $R = 3$ is used as a default.

TABLE VI

ABLATION RESULTS OF NUMBER OF SCALE HEADS

R	Dilation Rate	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	#Params	GFLOPs
2	$[1, 2]$	43.9	76.1	45.4	26.0	44.8	44.4	48.4M	62.6
3	$[1, 2, 3]$	44.8	77.4	46.4	21.7	48.8	45.9	50.3M	66.8
4	$[1, 2, 3, 4]$	42.9	76.8	44.6	16.5	46.0	45.9	52.2M	71.0

2) Effect of Scale-Agg: To assess the effect of Scale-Agg, the vanilla and split approaches are employed as alternatives to the Scale-Agg approach. For the vanilla approach, each scale head is directly operated on the input eight heads. The split approach means that the input eight heads are divided into three groups (used as three scale heads) in a 3:3:2 ratio. As evidenced in Table VII, the Scale-Agg approach outperforms the alternative variants, achieving the highest AP score.

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TABLE VII

ABLATION RESULTS OF SCALE-AGG. THE VANILLA APPROACH MEANS THAT EACH SCALE HEAD IS DIRECTLY OPERATED ON THE INPUT EIGHT HEADS. THE SPLIT APPROACH MEANS THAT THE INPUT EIGHT HEADS ARE DIVIDED INTO THREE GROUPS IN A 3:3:2 RATIO

Strategies	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	#Params	GFLOPs
Vanilla	44.0	76.5	47.0	16.3	48.3	46.6	50.3M	66.8
Split	38.8	66.9	39.7	18.1	36.5	45.3	46.0M	56.9
Scale-Agg	44.8	77.4	46.4	21.7	48.8	45.9	50.3M	66.8

TABLE VIII

ABLATION RESULTS OF DIFFERENT ATTENTION MECHANISMS. DIL-ATT AND DEF-ATT DENOTE DILATED ATTENTION AND DAT, RESPECTIVELY

Attention	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	#Params	GFLOPs
Dil-Att	31.4	64.0	26.7	7.6	36.4	36.6	45.5M	56.3
Def-Att	41.9	69.5	45.9	18.7	39.7	47.5	46.7M	58.7
SEDA	44.8	77.4	46.4	21.7	48.8	45.9	50.3M	66.8

3) Effect of SEDA: To evaluate the performance of SEDA, we compare it with dilated attention and DAT. Specifically, the OD approaches based on these three attention mechanisms are all implemented using the DINO framework. Apart from the attention blocks, these three approaches share identical implementation settings. Table VIII presents a quantitative comparison of different attentions. It can be observed that SEDA outperforms DAT and dilated attention in terms of AP.

V. CONCLUSION

In this article, we propose a novel end-to-end transformer-based RSOD method, called SED-DETR. The core component of SED-DETR is the SEDA module, designed to overcome the challenges caused by the diverse scales and shapes in RSOD. This module adaptively samples the points through multiscale dilation sampling and position updates with learnable offsets. By introducing the scale and shape prior to attention calculation, the SEDA module can improve the perceptibility of scale and shape. In addition, it employs a Scale-Agg approach to enhance the scale-aware modeling capability. The superiority of SED-DETR is verified on three remote sensing datasets (xView, DIOR, and NWPU VHR-10) and one generic natural image dataset (COCO). In future work, we will design task-specific object queries to improve SED-DETR further.

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